

MATH 231 – Linear Algebra I

The Vector Calculus Unit, Part 1 ■ Derivatives, Partial Derivatives, and the Gradient

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About this document. This opens a unit on calculus in several variables. It begins with a review of one-variable differentiation — material from MATH 141 — not as a courtesy, but because the central definition of §6 turns out to be the one-variable definition with a single substitution made inside it. Having that definition fresh, and in the same notation, is what makes the new one read as a small change rather than a new idea.

A common confusion (Section numbers restart). This unit numbers its sections from 1 again. They do *not* continue from the vector unit (§1–§24), the computational unit, or the matrix unit, all of which also start at 1. A bare “§5” below means §5 *of this unit*; whenever we point at another we will name it.

Assumed. From MATH 141: limits, and differentiation of one-variable functions. From the vector unit: $\mathbb{R}^n$ and its vectors (§24.1); function notation $f : A \rightarrow B$ (§6); the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ (§16, §24.1); the dot product (§11.3); and the Euclidean norm (§11.9).

What you should be able to do afterwards. State the definition of the derivative and use the standard rules; state the algebraic properties of e^x and $\ln x$ and explain why the two lists mirror each other; derive $\frac{d}{dx} \ln x$ from the definition and $\frac{d}{dx} e^x$ two ways; read the signature $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and give examples; define the partial derivative as a limit and compute one; say what kind of object a partial derivative is; assemble the partials of a function into its gradient; say which direction the gradient points and why that matters; write the linear approximation of a function of one variable and of n variables; build the Jacobian of a vector-valued function and the Hessian of a scalar one; state Clairaut’s theorem; and say why the Hessian is symmetric.

1. The derivative, in one variable

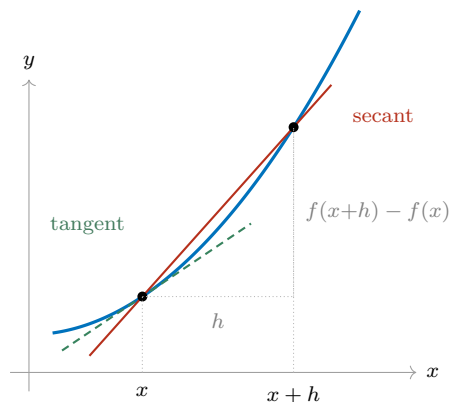
1.1. The definition

Definition 1.1 (Derivative). Let $f : \mathbb{R} \rightarrow \mathbb{R}$. The **derivative** of f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \tag{1}$$

provided the limit exists. When it does, f is called **differentiable** at x .

The quotient inside the limit is the slope of the line through the two points $(x, f(x))$ and $(x+h, f(x+h))$ — a *secant* line. As h shrinks, the second point slides toward the first and the secant pivots. The derivative is the slope it settles on: the slope of the tangent line at x , and the instantaneous rate at which f changes there.



1.2. Notation

The same object is written several ways, and all of them appear in practice:

$$f'(x), \quad \frac{df}{dx}, \quad \frac{dy}{dx}, \quad \frac{d}{dx}f(x).$$

The last form is useful because $\frac{d}{dx}$ reads as an *instruction* — “differentiate what follows with respect to x ” — which is how it is used in §4.2.

2. The rules

Computing every derivative from (1) would be unbearable, so the limit is evaluated once for each basic function and once for each way of combining functions. What follows is the resulting catalogue. Throughout, c is a constant and f, g are differentiable.

2.1. Structural rules

These say how differentiation interacts with the ways functions are built out of other functions.

Rule	Statement	Notes
Constant	$\frac{d}{dx} c = 0$	a flat graph has slope 0
Constant multiple	$\frac{d}{dx} [c f(x)] = c f'(x)$	scalars pass through
Sum / difference	$\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$	term by term
Power	$\frac{d}{dx} x^n = n x^{n-1}$	any real n
Product	$\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$	<i>not</i> $f'g'$
Quotient	$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$	order of the numerator matters
Chain	$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$	outer at inner, times inner

The first three together say that differentiation is *linear*: it distributes across sums and lets constants through. That word is worth registering — it is the same property, in the same sense, that the vector unit attached to other operations, and it will return when the gradient appears.

2.2. Trigonometric functions

Function	Derivative	Function	Derivative
$\sin x$	$\cos x$	$\csc x$	$-\csc x \cot x$
$\cos x$	$-\sin x$	$\sec x$	$\sec x \tan x$
$\tan x$	$\sec^2 x$	$\cot x$	$-\csc^2 x$

A pattern worth using as a memory aid: every co-function — $\cos$, $\csc$, $\cot$ — picks up a minus sign. The exponential and the logarithm are missing from these tables on purpose. Their derivatives are not quoted here; they are derived in §4.

3. The exponential and the logarithm

Before differentiating these two, their algebra is worth setting out — partly because §4.1 uses it, and partly because the two lists are the same list read in a mirror.

3.1. The exponential function

The function $\exp(x) = e^x$ has domain $\mathbb{R}$ and range $(0, \infty)$: it is *never* zero or negative, no matter what is fed to it. It is strictly increasing.

	Property	In words
E1	$e^0 = 1$	the exponential of nothing is one
E2	$e^{a+b} = e^a e^b$	sums become products
E3	$e^{a-b} = \frac{e^a}{e^b}$	differences become quotients
E4	$(e^a)^b = e^{ab}$	powers multiply the exponents
E5	$e^{-a} = \frac{1}{e^a}$	negation inverts

E2 is the essential one; E3 and E5 are consequences of it.

3.2. The natural logarithm

The function $\ln x$ has domain $(0, \infty)$ and range $\mathbb{R}$ — exactly the exponential's domain and range, exchanged. That is the first sign that the two are inverses.

	Property	In words
L1	$\ln 1 = 0$	mirrors E1
L2	$\ln(ab) = \ln a + \ln b$	products become sums
L3	$\ln \frac{a}{b} = \ln a - \ln b$	quotients become differences
L4	$\ln(a^r) = r \ln a$	exponents come down in front
L5	$\ln e = 1$	the base is its own unit

3.3. They are inverses

The two functions undo each other:

$$\ln(e^x) = x \quad \text{for all } x \in \mathbb{R}, \quad e^{\ln x} = x \quad \text{for all } x > 0. \quad (2)$$

This is what explains the parallel between the two tables. Read E1 through E4 backwards and they become L1 through L4: E2 says e turns addition into multiplication, so its inverse must turn multiplication back into addition, which is L2. The logarithm laws are not a second set of facts to memorise — they are the exponential laws seen from the other side.

A common confusion ($\ln$ of a sum). L2 converts the log of a *product* into a sum. There is no corresponding rule for the log of a sum: $\ln(a + b)$ does not simplify, and in particular it is not $\ln a + \ln b$. Check it once with $a = b = 1$: $\ln 2 \approx 0.693$, while $\ln 1 + \ln 1 = 0$.

4. Deriving the two derivatives

These two derivatives were deliberately left out of the tables in §2. Here they are earned instead. The order matters: the logarithm is done first, from the definition alone, and the exponential then follows from it.

4.1. The natural logarithm, from the definition

Let $f(x) = \ln x$ with $x > 0$. Apply (1) directly:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h}.$$

The numerator is a difference of logarithms, so L3 folds it into one:

$$= \lim_{h \rightarrow 0} \frac{1}{h} \ln\left(\frac{x+h}{x}\right) = \lim_{h \rightarrow 0} \frac{1}{h} \ln\left(1 + \frac{h}{x}\right).$$

Now use L4 in reverse — a coefficient in front of a logarithm can be pushed up into the exponent — moving the $1/h$ inside:

$$= \lim_{h \rightarrow 0} \ln\left[\left(1 + \frac{h}{x}\right)^{1/h}\right] = \ln\left[\lim_{h \rightarrow 0} \left(1 + \frac{h}{x}\right)^{1/h}\right].$$

Moving the limit inside $\ln$ is legitimate because $\ln$ is continuous, and a continuous function may be evaluated at the limit of its input.

The remaining limit is a standard one. Substituting $u = h/x$, so that $u \rightarrow 0$ as $h \rightarrow 0$ and $1/h = 1/(ux)$,

$$\lim_{h \rightarrow 0} \left(1 + \frac{h}{x}\right)^{1/h} = \lim_{u \rightarrow 0} \left[(1 + u)^{1/u}\right]^{1/x} = e^{1/x},$$

using $\lim_{u \rightarrow 0} (1 + u)^{1/u} = e$, which is one of the standard definitions of e . Therefore

$$f'(x) = \ln(e^{1/x}) = \frac{1}{x},$$

the last step by (2). ■

4.2. The exponential, by implicit differentiation

With $\frac{d}{dx} \ln x = 1/x$ in hand, the exponential costs three lines. Let

$$y = e^x.$$

Take the logarithm of both sides, which by (2) gives

$$\ln y = x.$$

Differentiate both sides with respect to x . The right side gives 1. The left side is a function of y , which is itself a function of x , so the chain rule applies:

$$\frac{d}{dx} [\ln y] = \frac{d}{dx} [x] \quad \implies \quad \frac{1}{y} \frac{dy}{dx} = 1.$$

Solving for the derivative and substituting back,

$$\frac{dy}{dx} = y = e^x. \quad \blacksquare$$

This technique — differentiating an equation in which y is not isolated, then solving for dy/dx — is **implicit differentiation**. It works here because writing $\ln y = x$ converts a hard derivative into an easy one at the cost of an extra chain-rule factor.

4.3. The exponential, from the definition

The same result comes out of (1). Let $f(x) = e^x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}.$$

By E2, $e^{x+h} = e^x e^h$, so e^x is a common factor of the numerator:

$$= \lim_{h \rightarrow 0} \frac{e^x e^h - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x (e^h - 1)}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h}.$$

The factor e^x does not involve h , so it passes outside the limit. What remains is the standard identity

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1, \tag{3}$$

which gives $f'(x) = e^x$. ■

Remark. It is worth being clear about what this second derivation does and does not establish. Setting $x = 0$ in (1) for $f(x) = e^x$ turns it into exactly (3) — so the identity *is* the statement that $f'(0) = 1$. The argument therefore does not conjure the derivative from nothing; it shows that the derivative everywhere follows from its value at the single point $x = 0$. That is a real statement, and a strong one: the exponential is pinned down completely by its slope at the origin. The derivation in §4.2 is the one that stands independently, resting on §4.1.

5. Functions of several variables

5.1. The signature

Everything so far has concerned $f : \mathbb{R} \rightarrow \mathbb{R}$ — one number in, one number out. Almost nothing worth computing has that shape.

From the vector unit. The notation $f : A \rightarrow B$ was fixed in §6 of the vector unit: the set before the arrow is what the function consumes, the set after it is what the function produces. That notation now does real work.

Definition 5.1 (Scalar function of several variables). A **scalar function of n variables** is a function

$$f : \mathbb{R}^n \longrightarrow \mathbb{R},$$

taking a vector $\mathbf{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$ and returning a single real number $f(\mathbf{x})$.

The input is a *vector*; the output is a *scalar*. That asymmetry is the whole subject of this section. Such functions are also called **scalar fields**: they assign one number to every point of a space.

5.2. Examples

Example 5.1. Each of the following is a function $\mathbb{R}^n \rightarrow \mathbb{R}$.

- (a) $f(x, y) = x^2 + y^2$, with $n = 2$. It returns the squared distance from the origin. $f(3, 4) = 25$.
- (b) $f(x, y) = xy$, with $n = 2$. The area of a rectangle with the given side lengths. $f(3, 4) = 12$.
- (c) $f(x, y, z) = x^2 + y^2 + z^2$, with $n = 3$. Three inputs, still one output.
- (d) $f(\mathbf{x}) = \|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$, for any n . *The Euclidean norm is a function $\mathbb{R}^n \rightarrow \mathbb{R}$* — it eats a vector and returns its length, a single number. You have been using an example of this kind of function since the vector unit’s §11.9.
- (e) $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x_1 + \dots + a_nx_n$ for a fixed vector $\mathbf{a}$. The dot product against a fixed vector is likewise $\mathbb{R}^n \rightarrow \mathbb{R}$ (vector unit §11.3).
- (f) $f(x, y) = e^{-(x^2+y^2)}$, a bump peaking at the origin and decaying in every direction.
- (g) The temperature at each point of a room, or the elevation at each point on a map: both assign one number to each point of a two- or three-dimensional space.

Remark. Examples (d) and (e) are worth a second look. Two of the most-used objects from the vector unit — the norm and the dot product — are exactly the kind of function this section is about. Nothing new was introduced; they were simply never described this way before.

5.3. What “the derivative” should mean here

Now try to differentiate one of these, and the definition (1) immediately fails to typecheck. It reads

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

and if the input is a vector then “ $x + h$ ” is a vector plus a number, which is undefined. Even if it were repaired, a deeper problem remains: for a one-variable function there are only two ways to leave the point x — left or right — and the two are related by a sign. In $\mathbb{R}^n$ there are infinitely many directions to leave $\mathbf{x}$ in, and f may change at a different rate along each one.

Walking east from a point on a hillside might take you uphill; walking north from the same point might take you down. “The rate of change of the elevation” is not yet a well-posed question. It becomes one only after a direction is named.

The next section names the simplest possible ones.

6. Partial derivatives

6.1. The definition

The simplest directions to move in are the coordinate directions, and the vector unit already has names for them.

From the vector unit. $\mathbf{e}_j$ is the j -th standard basis vector of $\mathbb{R}^n$: a 1 in slot j and 0 everywhere else (vector unit §16, §24.1). Adding $h\mathbf{e}_j$ to $\mathbf{x}$ therefore changes the j -th coordinate by h and leaves every other coordinate exactly where it was.

Definition 6.1 (Partial derivative). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The **partial derivative of f with respect to x_j** at the point $\mathbf{x}$ is

$$\frac{\partial f}{\partial x_j}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_j) - f(\mathbf{x})}{h}, \quad (4)$$

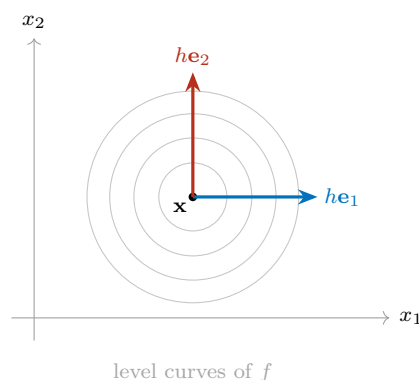
provided the limit exists.

Compare (4) with (1). They are the same limit. The only change is that the step h has become the step $h\mathbf{e}_j$ — a step of size h taken *in the j -th coordinate direction*. The curly ∂ replaces d to record that there were other directions available and this one was chosen.

Written out in coordinates for $n = 2$, with $\mathbf{x} = (x, y)$:

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}, \quad \frac{\partial f}{\partial y}(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}.$$

In the first, y never changes; in the second, x never changes.



Each arrow is one partial derivative: step off the point in a single coordinate direction, and measure the rate at which f responds.

6.2. Computing them

In practice the limit is never evaluated. Since every variable except x_j is held fixed, they behave as constants, and the expression becomes a function of the one live variable — so the rules of §2 apply unchanged.

To compute $\partial f/\partial x_j$: treat every variable other than x_j as a constant, and differentiate as in one-variable calculus.

Example 6.1. Let $f(x, y) = x^2y + 3y^2$.

For $\partial f/\partial x$, treat y as a constant. Then x^2y is a constant times x^2 , and $3y^2$ is a constant, whose derivative is zero:

$$\frac{\partial f}{\partial x} = 2xy.$$

For $\partial f/\partial y$, treat x as a constant. Now x^2y is a constant times y , and $3y^2$ is live:

$$\frac{\partial f}{\partial y} = x^2 + 6y.$$

Example 6.2. Let $f(x, y) = e^{-(x^2+y^2)}$, example (f) above. By the chain rule, holding y fixed,

$$\frac{\partial f}{\partial x} = e^{-(x^2+y^2)} \cdot (-2x) = -2x e^{-(x^2+y^2)}, \quad \frac{\partial f}{\partial y} = -2y e^{-(x^2+y^2)}.$$

Example 6.3. Let $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x_1 + \cdots + a_nx_n$, example (e) above. Only the j -th term involves x_j , and in it x_j appears to the first power with coefficient a_j ; every other term is a constant. So

$$\frac{\partial f}{\partial x_j} = a_j.$$

The partial derivatives of a dot product against $\mathbf{a}$ are the components of $\mathbf{a}$ itself.

6.3. What kind of object is a partial derivative?

This is worth stating carefully, because it is what the next section is built on.

Fix j . At each point $\mathbf{x} \in \mathbb{R}^n$, the limit (4) produces a single number. So $\partial f/\partial x_j$ assigns one real number to each point of $\mathbb{R}^n$ — which is to say it is itself a function

$$\frac{\partial f}{\partial x_j} : \mathbb{R}^n \longrightarrow \mathbb{R},$$

of exactly the same type as f . Differentiating with respect to one variable does not reduce the number of variables; the results above are visibly still functions of both x and y .

And there are n of them, one for each j from 1 to n .

7. The gradient

7.1. Collecting the partials

So a single function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ does not have a derivative. It has n of them — one per coordinate direction — and each is a function $\mathbb{R}^n \rightarrow \mathbb{R}$ in its own right. Carrying n separate functions around and naming them individually would be unmanageable for any interesting n , and it would obscure the fact that they belong together: they are n answers to the same question, asked in n directions.

From the vector unit. The vector unit’s opening move was exactly this. Faced with n related numbers, do not track them separately — package them into a single object of $\mathbb{R}^n$ and work with that. Position, velocity and force were treated that way; so were the coordinates of a vector against a basis.

Apply that move here.

Definition 7.1 (Gradient). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have all n partial derivatives at $\mathbf{x}$. The **gradient** of f at $\mathbf{x}$ is the vector in $\mathbb{R}^n$ whose j -th component is the j -th partial derivative:

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \frac{\partial f}{\partial x_2}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right].$$

The symbol ∇ is read “del” or “nabla”. Note the signature that results:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \text{but} \quad \nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n.$$

The gradient takes a point and returns a *vector*. This is a genuine departure from the one-variable case, where f' had the same signature as f .

7.2. Examples

Example 7.1. For $f(x, y) = x^2y + 3y^2$, the partials were computed above, so

$$\nabla f(x, y) = [2xy, x^2 + 6y].$$

At the point $(1, 2)$ this evaluates to $\nabla f(1, 2) = [4, 13]$ — a vector in $\mathbb{R}^2$ attached to that particular point. At a different point it is a different vector.

Example 7.2. For $f(x, y) = x^2 + y^2$, $\nabla f(x, y) = [2x, 2y] = 2[x, y]$. At every point the gradient points directly away from the origin.

Example 7.3. For $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle$, the last example of §6.2 gave $\partial f / \partial x_j = a_j$ for every j . Assembling them,

$$\nabla f(\mathbf{x}) = [a_1, a_2, \dots, a_n] = \mathbf{a}.$$

The gradient is $\mathbf{a}$ itself, the same at every point — the multivariable counterpart of the fact that a line of slope m has derivative m everywhere.

7.3. The direction it points

The gradient is a vector, so besides its components it has a *direction*, and that direction turns out to mean something. It is the single most useful fact about the gradient.

At a point $\mathbf{x}$, the gradient $\nabla f(\mathbf{x})$ points in the direction of **steepest ascent** of f — the direction in which f increases fastest. Its length $\|\nabla f(\mathbf{x})\|$ says how steep that climb is.

We will not prove this, but it should be made plausible, because it is the reason the gradient is worth computing at all.

Why it is believable. Stand somewhere on a hillside. Each partial derivative answers one narrow question: *if I walk due east, how fast does my elevation change?* And due north, and so on for each coordinate direction. A large positive $\partial f / \partial x_1$ means east is a good way to climb; a negative one means east goes downhill, so west climbs instead.

Now ask the real question — which way is uphill *fastest*? Intuitively you should take a bit of each direction that helps, and take more of the directions that help more. A step built by weighting each coordinate direction in proportion to how much it raises f is exactly a step of the form

$$\frac{\partial f}{\partial x_1} \mathbf{e}_1 + \frac{\partial f}{\partial x_2} \mathbf{e}_2 + \cdots + \frac{\partial f}{\partial x_n} \mathbf{e}_n,$$

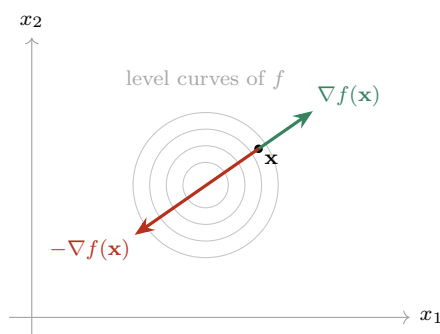
and that is the gradient. The directions that raise f most get the largest weights; the ones that lower it get negative weights, which points the step away from them.

From the vector unit. There is a second way to see it, using the vector unit. The rate of change of f in a chosen unit direction $\mathbf{u}$ turns out to be the dot product $\langle \nabla f, \mathbf{u} \rangle$, and by §11.6,

$$\langle \nabla f, \mathbf{u} \rangle = \|\nabla f\| \|\mathbf{u}\| \cos \theta = \|\nabla f\| \cos \theta,$$

since $\|\mathbf{u}\| = 1$. Among all unit directions this is largest when $\cos \theta = 1$, that is, when $\mathbf{u}$ points the same way as ∇f . The steepest climb is in the gradient's own direction, and the best possible rate is $\|\nabla f\|$ — which is why the length measures steepness.

The same thing happens in one variable. With $n = 1$ there are only two directions to move, and the gradient is the single number $f'(x)$. If $f'(x) > 0$ then f increases as x increases, so “uphill” is the positive direction; if $f'(x) < 0$, uphill is the negative direction. In both cases uphill is the direction the derivative points, and $|f'(x)|$ says how steep the slope is. The gradient is that statement with more than two directions available — which is why it must be a vector rather than a sign.



The two arrows show *directions*; their drawn lengths are not to scale. The gradient crosses the level curves at right angles, heading toward the ones where f is larger. Reversing it, $-\nabla f(\mathbf{x})$ gives the direction of steepest *descent* — here, straight toward the low point at the centre.

Looking ahead. That last observation is the whole idea behind an algorithm you will meet later. To find a point where f is as small as possible, start anywhere, compute $-\nabla f$, take a step in that direction, and repeat — walking downhill until the ground levels out. It is called *gradient descent*, and it is the workhorse of modern optimisation and machine learning. It exists because of the property in the box above.

7.4. Why this is the right package

Collecting the partials into a vector could look like mere bookkeeping — a convenient way to write n things at once. It is more than that, for a reason already visible in the last example.

The gradient of $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle$ came out as $\mathbf{a}$, not as a list of unrelated numbers that happen to be stored together. The package has an identity of its own. And because $\nabla f(\mathbf{x})$ is a vector in $\mathbb{R}^n$,

everything the vector unit built applies to it: it has a length, it has a direction, it can be dotted with other vectors, it can be projected, and it can be compared for orthogonality.

§7.3 has already spent two of those operations: the direction of ∇f and its length. The next section spends the dot product.

8. Linear approximation

One of the oldest uses of a derivative is to replace a complicated function, near a single point, by a simple one. The gradient does this in n dimensions exactly as f' does in one — and comparing the two formulas shows what kind of object the gradient really is.

8.1. One variable

Return to the picture in §1.1. The tangent line at $x = a$ touches the graph at that point and has slope $f'(a)$, so its equation is

$$L(x) = f(a) + f'(a)(x - a). \quad (5)$$

Near a the curve and its tangent line are hard to tell apart, so L can stand in for f :

$$f(x) \approx f(a) + f'(a)(x - a) \quad \text{for } x \text{ near } a.$$

This is the **linear approximation** (or *linearisation*) of f at a . Read (5) in three pieces: start from the known value $f(a)$, measure how far you have moved with $x - a$, and convert that displacement into a change in f by multiplying by the rate $f'(a)$.

Example 8.1. Estimate $\sqrt{4.1}$ without a calculator. Take $f(x) = \sqrt{x}$ and $a = 4$, where the value is known exactly. Then $f(4) = 2$ and $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$. Hence

$$\sqrt{4.1} \approx 2 + \frac{1}{4}(4.1 - 4) = 2 + \frac{0.1}{4} = 2.025.$$

The true value is $2.02484\dots$, so the estimate is off by about 1.5×10^{-4} — from arithmetic done in one line.

8.2. In n dimensions

Now let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and let $\mathbf{a} \in \mathbb{R}^n$ be the point we are near. Every piece of (5) survives, but two of them change type.

The starting value $f(\mathbf{a})$ is still a scalar. The displacement is now $\mathbf{x} - \mathbf{a}$, a *vector* in $\mathbb{R}^n$ — it must record how far we moved in every coordinate, not just one. And the rate is now $\nabla f(\mathbf{a})$, also a vector. To turn a vector of rates and a vector of displacements into a single number, there is exactly one natural operation available, and the vector unit built it: the dot product.

Definition 8.1 (Linear approximation in n variables). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have a gradient at $\mathbf{a}$. The **linear approximation** of f at $\mathbf{a}$ is

$$L(\mathbf{x}) = f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle, \quad (6)$$

and $f(\mathbf{x}) \approx L(\mathbf{x})$ for $\mathbf{x}$ near $\mathbf{a}$.

Set the two formulas side by side:

$$\underbrace{f(a) + f'(a)(x - a)}_{n=1} \quad \longleftrightarrow \quad \underbrace{f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle}_{\text{general } n}.$$

They are the same formula. The only change is that a product of two numbers has become a dot product of two vectors — and when $n = 1$ the dot product of two one-component vectors *is* ordinary multiplication, so (6) contains (5) as its smallest case.

Written out in coordinates, (6) reads

$$L(\mathbf{x}) = f(\mathbf{a}) + \frac{\partial f}{\partial x_1}(\mathbf{a})(x_1 - a_1) + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{a})(x_n - a_n),$$

one term per direction: each partial derivative converts the displacement in its own coordinate into a contribution to the change in f , and the contributions are added up.

Example 8.2. Let $f(x, y) = x^2 + y^2$ and approximate near $\mathbf{a} = (1, 2)$. There $f(\mathbf{a}) = 1 + 4 = 5$, and from §7.2, $\nabla f(x, y) = [2x, 2y]$, so $\nabla f(\mathbf{a}) = [2, 4]$. Then

$$L(x, y) = 5 + \langle [2, 4], [x - 1, y - 2] \rangle = 5 + 2(x - 1) + 4(y - 2).$$

At (1.1, 1.9) this predicts

$$L(1.1, 1.9) = 5 + 2(0.1) + 4(-0.1) = 5 + 0.2 - 0.4 = 4.8,$$

against the true value $1.21 + 3.61 = 4.82$. Note the two terms pulling opposite ways: moving right raises f , moving down lowers it, and the approximation adds both effects.

Remark (What the graph of L is). For $n = 1$ the linear approximation is a tangent *line*. For $n = 2$ it is a tangent *plane*, resting on the surface at one point. For general n it is a tangent hyperplane. In every case it is the flat object that best matches f at $\mathbf{a}$.

Remark (Why the gradient deserves to be called the derivative). (6) is the strongest answer to the question of what replaces f' in several variables. In one variable, $f'(a)$ is the number that makes the best linear approximation work. In n variables, $\nabla f(\mathbf{a})$ is the vector that plays that role — the same job, the same position in the formula, a different type of object. Collecting the partials into a vector was not bookkeeping after all: it produced the thing that behaves like a derivative.

Looking ahead. (6) is also where the next ideas come from. Keeping $\mathbf{x} - \mathbf{a}$ a fixed unit vector and reading off the resulting change gives the rate of f in that direction, completing the question left open in §5.3. Asking where L is flat — where $\nabla f(\mathbf{a}) = \mathbf{0}$ — locates the candidates for maxima and minima. And approximating with something better than a flat object requires a *second* derivative — which in n dimensions turns out to be a matrix rather than a vector. That matrix is the subject of §9.

9. The Jacobian and the Hessian

Differentiating $f : \mathbb{R}^n \rightarrow \mathbb{R}$ produced ∇f , which is not a function of the same type as f : it takes a point of $\mathbb{R}^n$ and returns a *vector* in $\mathbb{R}^n$. To differentiate a second time we first need to know what the derivative of such a function even is.

9.1. Functions with vector output

So far the output has always been a single number. Allow it to be a vector.

Definition 9.1 (Vector-valued function). A function $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ takes a vector $\mathbf{x} \in \mathbb{R}^n$ and returns a vector $\mathbf{F}(\mathbf{x}) \in \mathbb{R}^m$. Writing out the m components,

$$\mathbf{F}(\mathbf{x}) = [F_1(\mathbf{x}), F_2(\mathbf{x}), \dots, F_m(\mathbf{x})],$$

where each **component function** $F_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a scalar function of the kind §5 introduced.

So a vector-valued function is nothing more than m scalar functions travelling together — the same packaging move as §7.1, applied to outputs instead of derivatives.

Example 9.1. $\mathbf{F}(x, y) = [x^2 - y^2, 2xy]$ is a function $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, with $F_1(x, y) = x^2 - y^2$ and $F_2(x, y) = 2xy$. (If those two components look familiar, they are the real and imaginary parts of z^2 for $z = x + iy$.)

9.2. The Jacobian

Each component F_i has n partial derivatives. With m components that is mn numbers — too many for a vector, and naturally arranged in a grid.

Definition 9.2 (Jacobian). Let $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ have all partial derivatives at $\mathbf{x}$. Its **Jacobian** is the $m \times n$ matrix

$$\mathbf{J}_{\mathbf{F}}(\mathbf{x}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}, \quad (\mathbf{J}_{\mathbf{F}})_{ij} = \frac{\partial F_i}{\partial x_j}.$$

It is also written $D\mathbf{F}$, and it is what “the derivative of $\mathbf{F}$ ” means.

Read the shape carefully: **one row per output, one column per input**. Row i collects all the partials of the single component F_i — so row i is the gradient of F_i . The Jacobian is a stack of gradients, one for each component function.

When $m = 1$ there is a single row, and the Jacobian is the gradient written across. The gradient was the Jacobian all along, in the case where the output was one number.

Example 9.2. For $\mathbf{F}(x, y) = [x^2 - y^2, 2xy]$, differentiating each component with respect to each variable gives the 2×2 matrix

$$\mathbf{J}_{\mathbf{F}}(x, y) = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}.$$

Note that this matrix is *not* symmetric — $-2y$ against $2y$. Jacobians in general have no symmetry to them.

Example 9.3. For $\mathbf{F}(x, y, z) = [xyz, x + y + z]$, a function $\mathbb{R}^3 \rightarrow \mathbb{R}^2$, the Jacobian is 2×3 :

$$\mathbf{J}_{\mathbf{F}}(x, y, z) = \begin{bmatrix} yz & xz & xy \\ 1 & 1 & 1 \end{bmatrix}.$$

Two outputs, three inputs, so two rows and three columns.

9.3. The gradient is such a function

Now return to $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Section §7.1 recorded its gradient’s signature:

$$\nabla f : \mathbb{R}^n \longrightarrow \mathbb{R}^n.$$

That is exactly a vector-valued function of the kind just defined, with $m = n$. So it has a Jacobian — and that Jacobian is the second derivative we have been looking for.

9.4. The Hessian

Definition 9.3 (Hessian). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have all second partial derivatives at $\mathbf{x}$. Its **Hessian** is the $n \times n$ matrix of those second partials,

$$\mathbf{H}_f(\mathbf{x}) = \mathbf{J}_{\nabla f}(\mathbf{x}), \quad (\mathbf{H}_f)_{ij} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Written out for $n = 2$, with the variables called x and y :

$$\mathbf{H}_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x \partial x} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y \partial y} \end{bmatrix}.$$

The two entries on the diagonal are obtained by differentiating twice in the same variable. The two off the diagonal are the **mixed partials**: differentiate once in each variable, in the two possible orders.

Example 9.4. Take $f(x, y) = x^2y + 3y^2$, whose gradient was computed in §7.2:

$$\nabla f(x, y) = [2xy, \quad x^2 + 6y].$$

Differentiate each component with respect to each variable:

$$\frac{\partial}{\partial x}(2xy) = 2y, \quad \frac{\partial}{\partial x}(x^2 + 6y) = 2x, \quad \frac{\partial}{\partial y}(2xy) = 2x, \quad \frac{\partial}{\partial y}(x^2 + 6y) = 6,$$

so

$$\mathbf{H}_f(x, y) = \begin{bmatrix} 2y & 2x \\ 2x & 6 \end{bmatrix}.$$

The two mixed partials both came out $2x$, even though they were computed by differentiating in opposite orders.

Example 9.5. Take $f(x, y, z) = x^2y + y^2z + z^2x$, so that

$$\nabla f = [2xy + z^2, \quad x^2 + 2yz, \quad y^2 + 2zx].$$

Differentiating each component in each of the three variables gives

$$\mathbf{H}_f(x, y, z) = \begin{bmatrix} 2y & 2x & 2z \\ 2x & 2z & 2y \\ 2z & 2y & 2x \end{bmatrix}.$$

Again every mixed pair agrees: the $(1, 2)$ and $(2, 1)$ entries are both $2x$, the $(1, 3)$ and $(3, 1)$ entries are both $2z$, and the $(2, 3)$ and $(3, 2)$ entries are both $2y$.

Twice is a coincidence. It is not.

9.5. Clairaut's theorem

Theorem 9.1 (Clairaut). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. If the mixed second partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ exist and are continuous near a point, then at that point they are equal:*

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

In words: *the order of differentiation does not matter*. Differentiating first with respect to x and then with respect to y gives the same result as doing it the other way round. This is not obvious — the two computations pass through entirely different intermediate expressions, as both examples above show — and it is why the mixed entries kept agreeing.

A common confusion (The continuity hypothesis is doing work). Clairaut's theorem is a theorem, not a definition, and it can fail. There exist functions whose two mixed partials both exist at a point and are *not* equal there; what such functions lack is the continuity of those mixed partials near the point. Every function you will meet in this course — polynomials, exponentials, logarithms, sines and cosines, and anything built from them by the rules of §2 — satisfies the hypothesis comfortably, so the conclusion is always available. It is worth knowing that it rests on something.

9.6. The Hessian is symmetric

Clairaut's theorem is a statement about two numbers. Read as a statement about the matrix holding them, it says something clean.

The (i, j) entry of the Hessian is $\frac{\partial^2 f}{\partial x_i \partial x_j}$ and the (j, i) entry is $\frac{\partial^2 f}{\partial x_j \partial x_i}$. Clairaut says these are equal, for every pair i and j . So every entry of $\mathbf{H}_f$ equals its mirror image across the diagonal — which is precisely the definition of a symmetric matrix from the matrix unit, §6.5.

If the mixed second partial derivatives of f are continuous, then its Hessian is **symmetric**:

$$\mathbf{H}_f = \mathbf{H}_f^\top.$$

Both worked examples above are symmetric, as they had to be. Contrast this with the Jacobian of §9.2, which had no such property: a general Jacobian need not even be square, let alone symmetric. Symmetry is not a feature of second derivatives in general — it is a consequence of the Hessian being the second derivative of a *scalar* function, so that its two indices play interchangeable roles.

Remark. This is worth registering as more than a curiosity. Symmetric matrices are easier to work with than general ones in almost every respect, and a great deal of what can be said about maxima, minima and the shape of a surface near a point depends on the Hessian being symmetric. An $n \times n$ symmetric matrix also carries only $n(n+1)/2$ independent numbers rather than n^2 , so storing one costs roughly half as much.

Looking ahead. The gradient told us which way f is going; the Hessian tells us how that direction is itself changing — the curvature of f . Where the gradient vanishes, it is the Hessian that decides whether the flat point is a minimum, a maximum, or neither. And an optimisation method that uses curvature rather than slope alone can take much better-informed steps than the one built on the gradient by itself.